



University of Colombo, Sri Lanka
University of Colombo School of Computing
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester I - UCSC AY21 [held in Sept./ Oct. 2024]

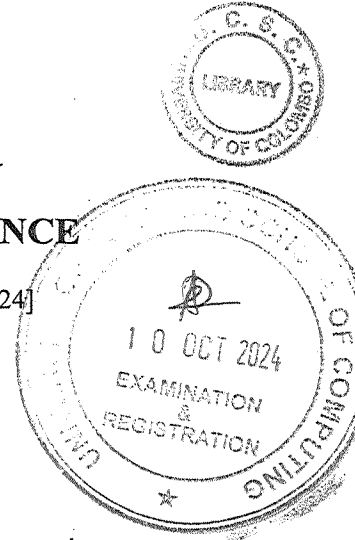
SCS 1302 – Discrete Mathematics

(Two (2) Hours)

Answer ALL questions

Number of Pages = 4

Number of Questions = 4



Important Instructions to candidates:

1. Students should answer in the medium of **English language only** using the answer paper provided.
2. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Write your index number **CLEARLY** on each and every page of the answer paper.
4. This paper consists of **4** questions in **04** pages (including the Cover Page).
5. Answer **ALL** questions.
6. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
7. **Non-Programmable** calculators are **not allowed**.

1.

- (a). Show that compound propositions $(p \wedge q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$ are logically equivalent by using a truth table.

[5 Marks]

- (b). Show that propositions $(p \vee q) \wedge \neg(\neg p \wedge q)$ and p are logically equivalent by developing a series of logical equivalences.

[5 Marks]

- (c). In the following program segment, n is an integer variable and A is an array $A[1], A[2], \dots, A[10]$ of 10 integer values.

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for  $n := 1$  to 10 do  
     $A[n] := n * n - n$ 
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Translate each of the following statements into logical expressions using quantifiers, and logical connectives. Let the domain consist of all the integers from 1 to 10, inclusive.

- (i). Every entry in the array is nonnegative.
- (ii). The entries in the array are sorted in strictly ascending order.
- (iii). The entries in the array are distinct.
- (iv). There exist two entries in A such that one is twice the other.

[8 Marks]

- (d). Test the validity of the following argument:

“First year students and third year students are not allowed to enroll in a graph theory course. Nimal is enrolled in a graph theory course. Therefore, Nimal is not a third-year student.”
Assume that the universe is the set of all Computer Science students at UCSC and Nimal is a particular Computer Science student at UCSC.

[7 Marks]

2.

- (a). Prove or disprove each of the following statements:

- (i). For all integers m and n , if mn is odd, then m and n are both odd.

[4 Marks]

- (ii). For every nonnegative integer n , $n^2 - n + 41$ is a prime number.

[3 Marks]

- (iii). If x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$, where $\max(x, y)$ is the maximum of x and y , and $\min(x, y)$ is the minimum of x and y .

[3 Marks]

(iv). There are irrational numbers x and y such that x^y is a rational number.

[4 Marks]

(b). Let A , and B be two subsets of a set $\mathcal{U}$. Define the following terms:

(i). Power set of A ($\wp(A)$); and

(ii). Symmetric difference of A and B ($A \oplus B$).

[2 Marks]

(c). Let A , B and C be three subsets of a set $\mathcal{U}$. Prove or disprove the following statements:

(i). $\wp(A \cap B) = \wp(A) \cap \wp(B)$.

(ii). $\wp(A \cup B) = \wp(A) \cup \wp(B)$.

(iii). If $A \oplus C = B \oplus C$, then $A = B$.

[9 Marks]

3.

(a). Consider the binary relation R on the set of integers ($\mathbb{Z}$) defined by $(a, b) \in R$ if and only if $ab \geq 1$ for $a, b \in \mathbb{Z}$.

Is R reflexive, symmetric, antisymmetric, and/or transitive? Justify your answer.

[8 Marks]

(b). Let S be a non-empty set. Show that the set inclusion relation " $\subseteq$ " is a partial ordering on the power set of S ($\wp(S)$). That is; $(\wp(S), \subseteq)$ is a partially ordered set (poset).

[4 Marks]

(c). Let $X = \{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}$. Answer the following questions for the poset $(X, \subseteq)$.

(i). Draw the Hasse diagram for the poset $(X, \subseteq)$.

(ii). Find the maximal elements of X .

(iii). Find the minimal elements of X .

(iv). Find the greatest element of X , if it exists.

(v). Find the least element of X , if it exists.

- (vi). Find all upper bounds of $\{\{2\}, \{4\}\}$.
- (vii). Find the least upper bound of $\{\{2\}, \{4\}\}$, if it exists.
- (viii). Find all lower bounds of $\{\{1, 3, 4\}, \{2, 3, 4\}\}$.
- (ix). Find the greatest lower bound of $\{\{1, 3, 4\}, \{2, 3, 4\}\}$, if it exists.

[13 Marks]

4.

- (a). Let a, b, c, d be integers and m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then show that $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

[6 Marks]

- (b). Find the value of $4^{100} \pmod{13}$.

[6 Marks]

- (c). State the Fermat's Little Theorem and use it to find the value of $7^{222} \pmod{11}$.

[6 Marks]

- (d). Find the greatest common divisor of 89 and 232 using the Euclidean algorithm. Hence, find the modular inverse of 89 modulo 232.

[7 Marks]